

Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference

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Abstract

Large Language Models (LLMs) have unlocked new capabilities and applications; however, evaluating the alignment with human preferences still poses significant challenges. To address this issue, we introduce Chatbot Arena, an open platform for evaluating LLMs based on human preferences. Our methodology employs a pairwise comparison approach and leverages input from a diverse user base through crowdsourcing. The platform has been operational for several months, amassing over 240K votes. This paper describes the platform, analyzes the data we have collected so far, and explains the tried-and-true statistical methods we are using for efficient and accurate evaluation and ranking of models. We confirm that the crowdsourced questions are sufficiently diverse and discriminating and that the crowdsourced human votes are in good agreement with those of expert raters. These analyses collectively establish a robust foundation for the credibility of Chatbot Arena. Because of its unique value and openness, Chatbot Arena has emerged as one of the most referenced LLM leaderboards, widely cited by leading LLM developers and companies. Our demo is publicly available at <https://chat.lmsys.org>.

1. Introduction

Recent advancements in large language models (LLMs) have significantly expanded their capabilities beyond traditional natural language processing boundaries, addressing a broad array of general tasks (OpenAI, 2023; Gemini et al., 2023; Touvron et al., 2023). These developments underscore the potential of LLMs but also have raised concerns with respect to performance evaluation. Current benchmarks often fail to capture the nuanced and diverse aspects of these models, particularly in assessing their alignment with human

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		Question Source	
Evaluation Metric	Ground Truth	Static	Live
	Human Preference	MMLU, HellaSwag, GSM-8K MT-Bench, AlpacaEval	Codeforces Weekly Contests Chatbot Arena

Figure 1. Classification of LLM benchmarks: We categorize along two dimensions: whether the questions are from a static dataset or a live, fresh source, and whether the evaluation metric relies on ground truth or (approximated) human preferences. MMLU (Hendrycks et al., 2020), HellaSwag (Zellers et al., 2019), GSM-8K (Cobbe et al., 2021), MT-Bench (Zheng et al., 2023b), and AlpacaEval (Li et al., 2023) are common examples of static benchmarks. Chatbot Arena is the platform introduced in this paper.

preferences in real-world, open-ended tasks.

To assess the performance of LLMs, the research community has introduced a variety of benchmarks. These benchmarks can be categorized based on two factors: the source of questions (either static or live) and the evaluation metric (either ground truth or human preference). According to these factors, benchmarks can be classified into four categories, as shown in Figure 1. While a range of benchmarks is beneficial, the most prevalent current method for evaluating LLMs remains a static, ground-truth-based evaluation, partly because such evaluations are inexpensive and reproducible.

However, these static, ground-truth-based benchmarks exhibit several limitations. Firstly, the questions within these benchmarks are not open-ended, hindering the ability to capture the flexible and interactive use found in real-world settings (Zheng et al., 2023b). Secondly, the test sets in these benchmarks are static, meaning they can become contaminated over time, which undermines the reliability of the evaluation results (Yang et al., 2023). Furthermore, for many complex tasks, establishing a definitive ground truth is not only challenging but sometimes unattainable. Consequently, current benchmarks fail to adequately address the needs of state-of-the-art LLMs, particularly in evaluating user preferences. Thus, there is an urgent necessity for an open, live evaluation platform based on human preference that can more accurately mirror real-world usage.

Creating such a benchmark platform entails significant challenges. It requires the collection of live, fresh, and diverse user questions to accurately represent real-world scenarios.

Additionally, developing scalable, incremental, and efficient ranking systems is essential for evaluating a large number of models. Moreover, ensuring the quality of human evaluations is crucial given the noisy nature of human preferences.

To this end, we introduce Chatbot Arena, a benchmarking platform for LLMs that features anonymous, randomized battles in a crowdsourced setting. Chatbot Arena is a free website open to all users.¹ On this website, a user can ask a question and get answers from two anonymous LLMs. Afterward, the user casts a vote for the model that delivers the preferred response, with the models’ identities revealed only after voting. This crowdsourced method effectively gathers a diverse array of fresh user prompts, accurately reflecting real-world LLM applications. Armed with this data, we employ a suite of powerful statistical techniques, ranging from the statistical model of Bradley & Terry (1952) to the E-values of Vovk & Wang (2021), to estimate the ranking over models as reliably and sample-efficiently as possible. With these tools in hand, we have designed efficient sampling algorithms specifically to select model pairs in a way that accelerates the convergence of rankings while retaining statistical validity.

We conduct a thorough analysis of the collected data to ensure the credibility of our platform. We demonstrate that the user-generated questions are sufficiently diverse to encompass a wide range of LLM use cases and are sufficiently challenging to differentiate between models. Furthermore, we confirm that the crowd-sourced votes are highly consistent with expert evaluations.

We have been running our system since Apr 2023 and have received over 240K votes from about 90K users in over 100 different languages as of Jan 2024. To encourage user engagement, we have made over 50 state-of-the-art models available for free. We also collaborate with leading model developers such as OpenAI, Google, Anthropic, Mistral, Hugging Face, and various universities, incorporating their latest models into our platform. We keep the community engaged by routinely updating the leaderboard, publishing analytical blogs, releasing datasets, and sharing information via tweets. Because of its unique and significant value, our leaderboard has emerged as one of the most referenced in the LLM field and has become a benchmark for the industry. We commit to making our data and code available, ensuring that this platform is open-source and open-accessible.

We make the following contributions:

- We build the first large-scale crowd-sourced live LLM evaluation platform with over 1M users visit.²

¹<https://chat.lmsys.org>

²The number was estimated by Google Analytics as of March 2024. Note that user visit may not convert to votes as our website also offers “direct chat” mode.

- We conduct an in-depth analysis of the collected data, including prompt diversity, quality, vote quality, and insights on human feedback.
- We will publicly release a human preference dataset with over 100K pairwise votes collected from Chatbot Arena.
- We design an efficient sampling algorithm that actively chooses which model pairs to show, such that our sample efficiency improves, sometimes to a large degree.

2. Related Work

LLM Benchmarks. We briefly review the common LLM benchmarks, following the classification presented in Figure 1. The most prevalent benchmarks are static, ground-truth-based ones, typically in the form of multiple-choice questions or question-answering tasks with predefined answers and test cases. These benchmarks encompass a range of topics including language understanding, mathematics, coding, and logical reasoning. Prominent examples in this category are MMLU (Hendrycks et al., 2020), Hel-laSwag (Zellers et al., 2019), GSM-8K (Cobbe et al., 2021), BigBench (Srivastava et al., 2023), AGIEval (Zhong et al., 2023), and HumanEval (Chen et al., 2021). Benchmarks focusing on safety, such as ToxicChat (Lin et al., 2023), and comprehensive suites like HELM (Liang et al., 2022), also exist. In addition to closed-ended questions, benchmarks can include open-ended questions that are evaluated by human judgment, which can be rated by experts or crowd workers such as Amazon Mechanical Turk (Karpinska et al., 2021; Geng et al., 2023; Wang et al., 2023). The recent trend includes utilizing GPT-4 for approximating human judgment (Chiang & Lee, 2023), with notable instances being MT-Bench (Zheng et al., 2023b) and AlpacaEval (Li et al., 2023). In addition to static benchmarks, live benchmarks that include fresh questions are also available. These questions can be obtained from annual exams or weekly online contests such as Codeforces (Li et al., 2022; Huang et al., 2023). They can also be sourced from human interaction. Some studies have explored using live human interaction for reinforcement learning from human preference (Bai et al., 2022; Ouyang et al., 2022; Touvron et al., 2023). However, these studies are typically limited to specific organizations. In this paper, we introduce Chatbot Arena, the first open, large-scale, and crowdsourced benchmark platform that utilizes live human interaction.

Risks of Static Benchmarks. Static benchmarks have certain issues, including contamination, saturation, overfitting, and a lack of human alignment (Yang et al., 2023; Oren et al., 2023). DynaBench (Kiela et al., 2021) identifies these challenges and recommends the use of a live benchmark that incorporates a human-in-the-loop approach for classical NLP benchmarks. Our system adopts a similar spirit. However, our focus is on chatting with LLMs, and we implement

this on a significantly larger user scale.

Ranking System. Ranking systems have been a well-studied topic in statistics. Related topics include probability models (Hunter, 2004; Rao & Kupper, 1967), rank elicitation (Szörényi et al., 2015; Busa-Fekete et al., 2014a;b), and online experiment design (Chernoff, 1992; Karimi et al., 2021). The Elo rating system has also been used for LLMs (Bai et al., 2022; Boubdir et al., 2023). Contributing to this literature, we introduce techniques for accelerating ranking convergence and detecting abnormalities, specifically applied to large-scale, real-world settings of LLMs.

Human Preference Dataset. Owing to the significance of human preferences, several datasets and analyses exist that incorporate human preferences. These include OpenAssistant (Köpf et al., 2023), HH-RLHF (Bai et al., 2022), LMSYS-Chat-1M (Zheng et al., 2023a), and synthetic approximations of human preferences like UltraFeedback (Cui et al., 2023) and Nectar (Zhu et al., 2023). Our prior data release, LMSYS-Chat-1M (Zheng et al., 2023a), is similarly collected via crowdsourcing. However, LMSYS-Chat-1M comprises solely conversations and lacks human preference data, rendering it unsuitable for direct use in ranking studies. This paper focuses on the analysis of preference data for ranking purposes.

3. Human Preference Data Collection

In this section, we discuss our interface design to collect human preferences and present summary statistics.

3.1. Interface

Chatbot Arena crowd-sources feedback from users for model evaluation. Our goal is to design an ease-of-use interface to reduce friction for users to contribute data. Since we collect feedback from many users, it is difficult to set a consistent grading rubric across different people. Hence, we adopt a pairwise comparison mechanism where users only need to compare two model responses and vote for the better one, instead of requiring users to provide an absolute score.

In each battle, two anonymous models are sampled. To encourage data diversity, we do not preset any input prompt on the website. Users are free to input any prompt to the two models. We believe this creates incentives for user engagement, particularly given that we offer a free service. It also helps us collect a diverse set of inputs representing real-world usage. After models provide their answers, user compare them side-by-side and vote for the preferred answer. If a user cannot choose in the first turn, the user can continue chatting until identifying a winner. For those who are unsure, we also present two buttons, “tie” or “both are bad.” Figure 8 shows a screenshot of our interface. Before

using our service, users are required to accept terms of use, which gives us their consent to release the data publicly.

3.2. Data Statistics

We began collecting data in April 2023. As of Jan 2024, we have received around 240K votes from over 90K users. Our data involves more than 50 models, including both proprietary models like GPT-4, Claude, and Gemini, as well as open models such as LLaMA and Mistral. These conversations cover more than 100 languages, with 77% being in English, 5% in Chinese, and the remaining languages, such as Russian, German, Spanish, French, and Japanese, each representing less than 2% of the total. Each data point includes multi-turn conversations between the user and two LLMs, and a vote to indicate which model the user prefers. We summarize statistics in Table 1 along with other existing human preference datasets.

Figure 10 in the Appendix shows the vote count per model. On average, 8K votes are collected for each model. In Figure 2, we select a set of representative models and present their win rate and the number of battles. Note that we employ non-uniform sampling to concentrate votes on model pairs that have similar performance due to higher uncertainty. This helps us reduce the number of votes required to reach stable results. We later develop an adaptive sampling method and demonstrate its effectiveness against random sampling. See Section 5 for further analysis.

To ensure anonymity, we use keywords to filter out conversations containing model identity such as model name (e.g., GPT, Claude) or companies (e.g., OpenAI, Anthropic). To avoid misuse, we adopt OpenAI moderation API to flag conversations that contain unsafe content. The flagged user requests account for 3% of the total requests. Figure 9 in the Appendix shows the number of valid user votes over time, where we get 1-2K votes per day in recent months and spikes as we introduce new models or leaderboard updates.

4. From Pairwise Comparisons to Rankings

Our data consists of pairwise comparisons—but how can we use these comparisons to recover a ranking over all M models? This is a well-studied topic in the literature on learning to rank (Liu et al., 2009), and we present our perspective here. We let $\mathcal{A} = \{(m, m') : m < m' \text{ and } m, m' \in [M]\}$ denote our comparative data set.

We consider a sequential setting, where at time $t \in \mathbb{N}$, we serve the human a pair of models $A_t \in \mathcal{A}$ (which we pick), and in turn we observe the human’s response $H_t \in [0, 1]$. As an example, we might have that $A_t = (1, 2)$ and $H_t = 1$, indicating that the human prefers model 2 over model 1. In the ensuing text, we will primarily focus on the binary case—where $H_t \in \{0, 1\}$ —but our approach will generalize to

Table 1. Statistics of human preference datasets, including Anthropic HH (Bai et al., 2022), OpenAssistant Conversations (Köpf et al., 2023), and Chatbot Arena (as of 2024/1/21). The tokens are counted by Llama2’s tokenizer. “Conv” = Conversation. “Lang” = Language.

Dataset	# Convs	# Models	# Users	# Langs	Avg. # Turns per Sample	Avg. # Tokens per Prompt	Avg. # Tokens per Response
Anthropic HH	338,704	-	143	1	2.3	18.9	78.9
OpenAssistant	66,497	-	13,500	35	-	36.9	214.2
Chatbot Arena (20240121)	243,329	50	90,051	149	1.3	94.9	269.0

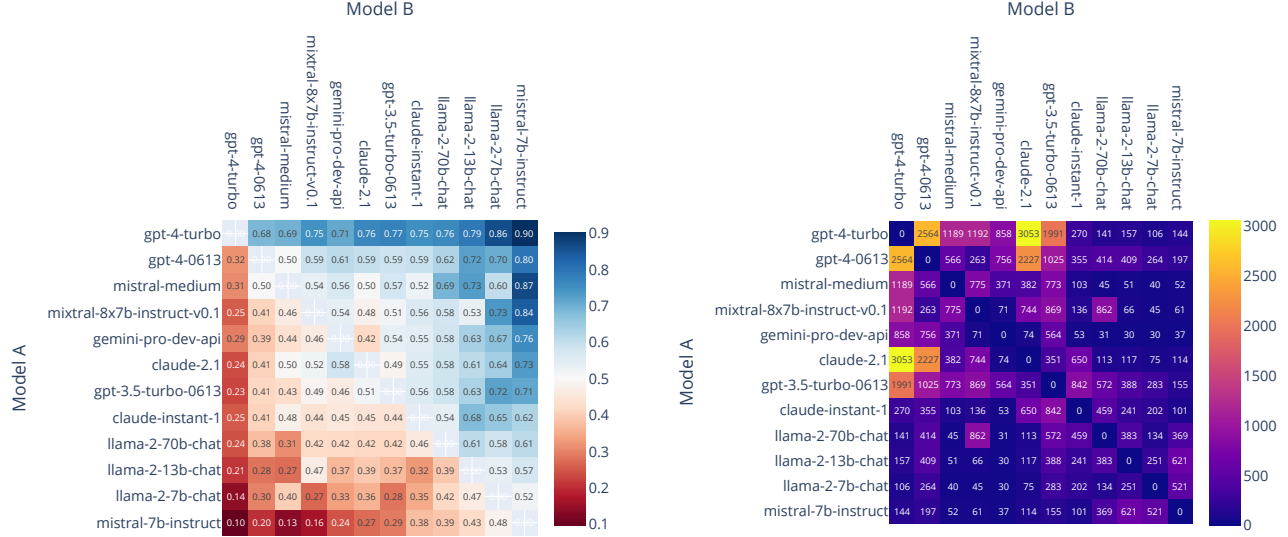


Figure 2. Win-rate (left) and battle count (right) between a subset of models in Chatbot Arena.

any form of feedback, including the possibility of allowing the human to express different degrees of preference or to say the models are tied.

One critical goal is to estimate the *win matrix*: $\theta^*(a) = \mathbb{E}[H_t \mid A_t = a]$, for all $a \in \mathcal{A}$; see the left panel of Figure 2 for an illustration of the (empirical) win matrix. In the binary case, the a entry in the win matrix corresponds to the probability the human prefers model a_2 to a_1 when shown the pair a . Finding the win matrix is a relatively straightforward mean-estimation problem; we will provide details in Section 5.

Formally, consider a *score* $s(\mathbb{P}) \in \mathbb{R}^M$, where $\mathbb{P}$ is a joint distribution over A and H (by default, we will target a uniform distribution over $\mathcal{A}$). Each model has a true score $s(\mathbb{P})_m$, and better models will have higher scores. In particular, we have the rank of model m :

$$\text{rank}(\mathbb{P})_m = 1 + \sum_{m' \in [M]} \mathbb{1}\{s(\mathbb{P})_{m'} > s(\mathbb{P})_m\}. \quad (1)$$

The best model has rank 1. If there is another model tied for best, they will both get assigned rank 1.

Picking a score. A standard score function in this setting is the vector of Bradley-Terry (BT) coefficients (Bradley & Terry, 1952). In the Bradley-Terry model, $H_t \in \{0, 1\}$, and the probability model m beats model m' is modeled via a

logistic relationship:

$$\mathbb{P}(H_t = 1) = \frac{1}{1 + e^{\xi_{m'} - \xi_m}}, \quad (2)$$

where ξ is an M -length vector of so-called BT coefficients. Without loss of generality, we take $\xi_1 = 0$ (since the model is invariant to addition in ξ). Our goal is to estimate the population Bradley-Terry coefficients, i.e., those that minimize the binary cross-entropy:

$$s(\mathbb{P}) = \underset{\xi}{\operatorname{argmin}} \mathbb{E}_{(A,H) \sim \mathbb{P}} \left[\ell \left(H, \frac{1}{1 + e^{\xi_{A_2} - \xi_{A_1}}} \right) \right], \quad (3)$$

where ℓ is the binary cross-entropy loss, $\ell(h, p) = -(h \log(p) + (1 - h) \log(1 - p))$.

Although the BT model technically assumes a parametric form for the model win rates, the seminal results of Huber et al. (1967); White (1982) show that maximum likelihood estimators are still asymptotically normal even when these assumptions *do not* hold, so long as the so-called “sandwich” covariance matrix is used; see Section 5 for details, and see Appendix B for a nonparametric extension of the Bradley-Terry model. Finally, we remark that previous evolutions of our online interface have reported different ranking scores, such as the Elo score (Elo, 1967) instead of the BT coefficients. We made this change because the BT coefficients are better for the purpose of statistical estimation.

5. Efficient Approximate Ranking

In Section 4 we described how to calculate the win matrix, score, and rank. Now we describe our estimation procedures.

Win matrix estimation. Estimation of the win matrix is relatively straightforward. Define $X_t(a) = \frac{1}{P_t(a)} H_t \mathbb{1}\{A_t = a\}$, where $P_t(a)$ is the probability of sampling pair a at time t , and X_t as the according vector. Then the estimator is

$$\hat{\theta}_T = \frac{1}{T} \sum_{t=1}^T X_t. \quad (4)$$

Note that $\mathbb{E}[X_t(a)] = \theta^*(a)$ for all t , and thus $\hat{\theta}_T$ is an unbiased estimator of θ^* . We will furthermore estimate the covariance matrix as

$$\hat{\Sigma}_T = \frac{1}{T} \sum_{t=1}^T (X_t - \hat{\theta}_T)(X_t - \hat{\theta}_T)^\top. \quad (5)$$

Under the appropriate regularity conditions, we have that

$$\sqrt{T} \hat{\Sigma}^{-1/2}(\hat{\theta} - \theta^*) \rightarrow \mathcal{N}(0, I_d), \quad (6)$$

and we construct confidence intervals accordingly. For an understanding of the appropriate regularity conditions, see Durrett (2019), Theorem 8.2.8, where condition (ii) is trivially satisfied so long as $P_t(a) > \epsilon > 0$, and condition (i) is implied by the almost-sure convergence of $P_t(a)$ to a limiting distribution $P(a)$.

Estimating the BT scores. To estimate the BT coefficients, mirroring (3), we perform (reweighted) maximum likelihood estimation on our data points:

$$s(\hat{\mathbb{P}}) = \underset{\xi}{\operatorname{argmin}} \sum_{t=1}^T \frac{1}{P(A_t)} \ell \left(H_t, \frac{1}{1 + e^{\xi_{A_t,2} - \xi_{A_t,1}}} \right), \quad (7)$$

where $A_t \sim P$. We perform the inverse weighting by $P(A_t)$ because this allows us to target a score with a uniform distribution over A .

To compute confidence intervals on the BT coefficients, we employ two strategies: (1) the pivot bootstrap (DiCiccio & Efron, 1996), and (2) the “sandwich” robust standard errors outlined in Huber et al. (1967) (see also Freedman (2006) for an outline of the necessary technical assumptions). Ultimately, based on the results of a simulation study described in Appendix A, we choose to deploy the sandwich intervals due to their smaller size in large samples.

Approximate rankings. Finally, we report an approximate ranking for each model that accounts for the uncertainty in the estimation of the score. Given an M -dimensional confidence set $\mathcal{C}$ satisfying

$$\mathbb{P}(s(\mathbb{P}) \in \mathcal{C}) \geq 1 - \alpha, \quad (8)$$

we extract an approximate ranking $R_m = 1 + \sum_{m' \in [M]} \mathbb{1}\{\inf \mathcal{C}_{m'} > \sup \mathcal{C}_m\}$. The uniform validity of $\mathcal{C}$ directly implies that $\mathbb{P}(\exists m : R_m > \operatorname{rank}(\mathbb{P})_m) \leq \alpha$ —i.e., with high probability, no model’s performance is understated. A guarantee on the other side—that no model’s performance is overstated—is possible by interchanging the inf and sup. To get the uniform confidence set, we construct the chi-squared interval implied by the central limit theorem using the sandwich estimate of the variance. In other words, we construct the interval $\{\xi : T \|\hat{V}^{-1/2}(\hat{\xi} - \xi)\| \leq \chi_{1-\alpha, M-1}^2\}$, where $\hat{\xi}$ is our MLE of the BT coefficients and $\hat{V}_\xi$ is the sandwich variance of the logistic regression.

Active sampling rule. Our sampling rule was to choose the model pair $a \in \mathcal{A}$ proportionally to the reduction in confidence interval size by sampling that pair:

$$P_t(a) \propto \sqrt{\frac{\hat{\Sigma}_{t,a,a}}{|\{t : A_t = a\}|}} - \sqrt{\frac{\hat{\Sigma}_{t,a,a}}{|\{t : A_t = a\}| + 1}}. \quad (9)$$

5.1. Detecting Anomalous Users

On a different note, we take a first step towards identifying anomalous IP addresses in our dataset. In a dataset of U unique IPs, we let $\text{IP} = \{1, \dots, U\}$ be the set of all IP addresses. Consider a “test” user, outside this database, who gives ratings $H'_1, \dots, H'_n$ when presented actions $A'_1, \dots, A'_n$. The idea of our procedure is to compare the distribution of ratings for the new user to the historical distribution of ratings for a given action. We let $\mathcal{H}_a = \{H_t : A_t = a\}$ and every time a user submits a vote, we calculate the following number:

$$p_i = \frac{1}{|\mathcal{H}_{A'_i}| + 1} \left(1 + \sum_{h \in \mathcal{H}_{A'_i}} \mathbb{1}\{h \geq H'_i\} \right). \quad (10)$$

Under the null hypothesis that $\mathcal{H}_{A'_i}$ is exchangeable with H'_i , p_i is a valid p-value (see Appendix C for a proof). Furthermore, the dependence of these p-values asymptotically is negligible.

With this p-value in hand, we can test against this null hypothesis sequentially by using Fisher’s combination test (Fisher, 1928) along with a variant of the Bonferroni correction. In particular, for each user, after their j th vote, we compute $M_j = -2 \sum_{i=1}^j \log(p_i)$. At 5 randomly chosen values of j between 1 and 100, we identify a user as anomalous if $M_j \geq \chi_{2j, 1-\alpha/5}^2$. (The times are randomly chosen, as to avoid anomalous users strategizing to hack this p-value.) Despite the heuristic application of this procedure, it seems to work well in our small-scale tests reported in Table 5.

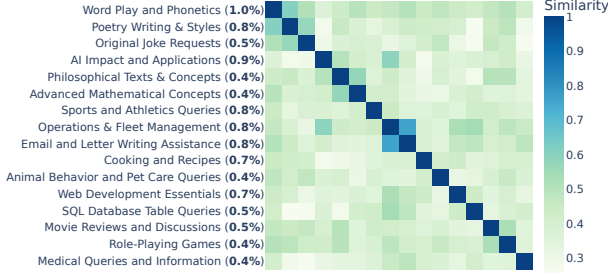


Figure 3. Similarity matrix of top-16 topic clusters. The number followed by the topic label represents the cluster size in percentage. Note that similarity is computed by cluster’s centroid embeddings, hence diagonals are always one.

6. Data Analysis

To examine whether Arena’s crowdsourced data reflects real-world use cases, we conduct topic modeling on the user prompts. We show how effective are these prompts in distinguishing models. Lastly, we validate the vote quality by relabeling data with experts.

6.1. Topic Modeling on User Prompts

To study the prompt diversity, we build a topic modeling pipeline with BERTopic³ (Grootendorst, 2022). We start with transforming user prompts into representation vectors using OpenAI’s text embedding model (text-embedding-3-small). To mitigate the curse of dimensionality for data clustering, we employ UMAP (Uniform Manifold Approximation and Projection) (McInnes et al., 2020) to reduce the embedding dimension from 1,536 to 5. We then use the hierarchical density-based clustering algorithm, HDBSCAN, to identify topic clusters with minimum cluster size 32. Finally, to obtain topic labels, we sample 10 prompts from each topic cluster and feed into GPT-4-Turbo for topic summarization.

The pipeline identifies 600 clusters covering a wide range of topics including poetry writing, coding, math, and medical queries. We present the top-16 topic clusters in Figure 3. We observe that the largest cluster only accounts for 1% of the entire set and the rest quickly drop to <0.5%, and the similarity between clusters is small, showing a long-tail and diverse distribution. Due to space limit, we present the similarity matrix and cluster hierarchy of top-64 clusters in Figure 11 and 12 in Appendix.

6.2. Can Arena Prompts Distinguish Models?

Next, we study how effective are these topic clusters in distinguishing models strengths. Constructing challenging prompts has become increasingly difficult due to LLMs’

Table 2. GPT-4-0613’s win-rate against Llama-2-70b-chat on 30 sample prompts from various topic clusters. We use GPT-4-turbo as judge to evaluate model responses in pairwise comparison.

Topic Cluster	Win-rate	Size
Python Game Programming Challenge	96.7%	0.2%
C/C++ Process Multi-Threading	86.7%	0.3%
SQL Query Database Assistance	73.3%	0.2%
Poetry Writing Prompts	66.7%	1.1%
Python Coding Basics	65.0%	0.2%
Linguistic Analysis & Wordplay	58.3%	0.7%
Travel Itinerary Planning	58.3%	0.4%
Movie Recommendations & Ratings	53.3%	0.2%

fast growing capabilities. For example, open models such as Llama-2-70b-chat can likely answer inquiries about movie or travel recommendation as good as GPT-4, but not in other domains such as reasoning or coding. To demonstrate, we sample 30 prompts from seven topic clusters and compare the performance of Llama-2-70b-chat and GPT-4. To control variables, we factor out user votes and consider LLM-as-judge (Zheng et al., 2023b) to evaluate model response. Results are shown in Table 2, where we see GPT-4 has significantly higher win-rate (up to 97%) in clusters that require coding and reasoning skills. On the other hand, for clusters with less problem-solving tasks, GPT-4 win-rate drops to below 60%. We show examples in Appendix D.1. This result shows models may exhibit varying strengths in different areas, but also highlights some of the topic clusters in Chatbot Arena are effective in differentiate models.

Building Challenging Benchmark. To further demonstrate the prompt quality, we show it is possible to construct a challenging benchmark with crowd-sourced user prompts. To ensure both topic coverage and quality, we first run the topic modeling pipeline and follow a similar procedure in Zheng et al. (2023a) to select challenging questions sampled from each topic cluster. Examples prompts and evaluation procedures can be found in the Appendix D.2 and Appendix D.3, respectively. We observe the selected prompts are highly effective in differentiating models. In Figure 4, we compare Arena bench against a widely used LLM benchmark, MT-Bench (Zheng et al., 2023b). We can see that Arena Bench effectively reveals a significant gap in performance between proprietary and the strongest open models.

6.3. Validating Vote Quality

To assess the quality of crowdsourced votes, we randomly selected 160 battles between GPT-4-Turbo and Llama-2-13B, as well as GPT-4-Turbo and GPT-3.5-Turbo-0613. We then asked experts⁴ to label their preference per comparison. The experts were given the prompts and answers blindly, and asked to carefully fact-check model’s answer with external resources like search engine. Manually labeling each

³<https://github.com/MaartenGr/BERTopic>

⁴The laborers are graduate students at UC Berkeley.

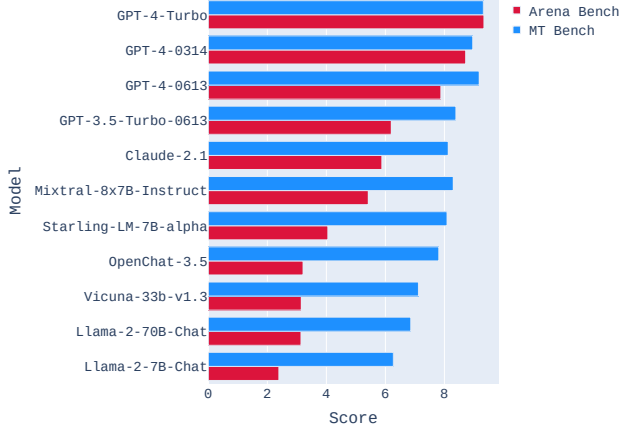


Figure 4. Model’s performance between Arena Bench and MT-Bench, showing an increased gap between open and proprietary models. Both uses GPT-4 as judge.

Table 3. Pairwise agreement rate between crowd-user, gpt-4 judge, and experts on pairwise battles. The top part of the table is between GPT-4-Turbo and Llama-2-13b-chat. The bottom is between GPT-4-Turbo and GPT-3.5-Turbo-0613.

Llama-2-13b	Expert 1	Expert 2	GPT-4
Crowd	72.8%	77.8%	75.6%
Expert 1	-	89.8%	81.0%
Expert 2	-	-	78.5%
GPT-3.5-Turbo	Expert 1	Expert 2	GPT-4
Crowd	73.8%	83.1%	75.6%
Expert 1	-	79.4%	76.3%
Expert 2	-	-	79.3%

data point took on average 3-5 minutes. For reference, we also use GPT-4 as a judge for pairwise comparisons. The agreement rate between crowd-users, experts, and GPT-4-judge are presented in Table 3. The corresponding win-rate are shown in Table 4.

To summarize, we observe high agreement rates (72% to 83%) between Arena crowd-user and experts in both setup. Note that agreement rates between two experts are around similar levels (79.4% and 89.8%). As for the 10%-20% disagreement between experts, it is mostly due to some user prompts don’t have a ground truth answer. Depending on the preference of the evaluator, sometimes both answers can be argued as being better than the other one, such as the examples in Appendix D.4. The gap between crowd-vs-expert agreement rate and expert-vs-expert agreement rate (5%-10%) is mostly attributed to crowd user making mistakes or overlooking factual errors in model’s response. Overall, the agreement rates presented in Table 3 validate the decent quality of crowd-sourced votes in Chatbot Arena.

Table 4. GPT-4-Turbo’s win-rate across crowd-user, gpt-4 judge, and experts on pairwise battles against Llama-2-13b and GPT-3.5-Turbo-0613.

Baseline	Arena User	Expert 1	Expert 2	GPT-4
Llama-2-13b	81.2%	89.4%	86.9%	78.8%
GPT-3.5-Turbo	76.3%	82.5%	89.4%	79.4%

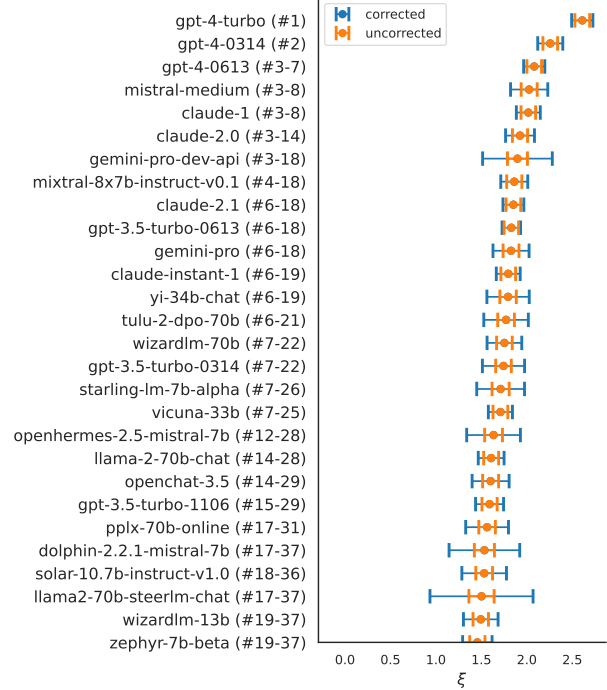


Figure 5. Intervals for the BT coefficients with and without multiplicity correction. The multiplicity correction, in this case a chi-square CLT interval, is technically required for the purpose of calculating the ranking, because it ensures *all* scores are simultaneously contained in their intervals (and the ranking is a function of all the scores). However, it induces extra conservatism, so we report both intervals.

7. Experiments

7.1. Ranking system

Computing the rank on real data. In this section, we report results from our experiments on approximate ranking. For this experiment, we ran a replay of $T = 213, 576$ historical votes from our online platform and calculate the BT coefficients using our earlier-described estimation algorithm with confidence intervals; see Figure 5 for these intervals (with and without multiplicity correction; the formal notion of approximate ranking technically requires multiplicity correction, but it makes the intervals looser).

Evaluating the coverage of the intervals. A natural follow-up question is whether or not the intervals are doing their job correctly: whether they cover the true BT coefficients with probability at least (and almost exactly) $1 - \alpha$. Of course,

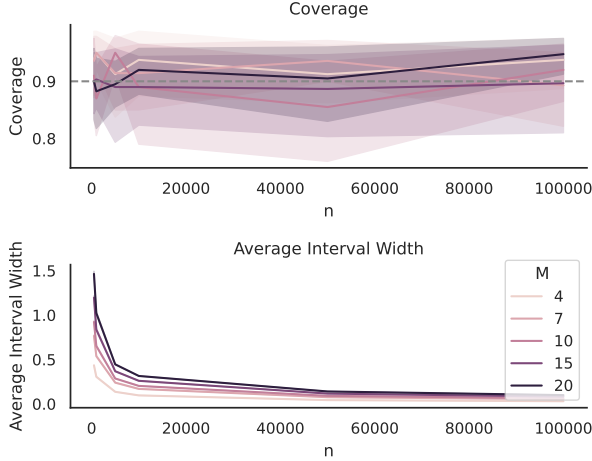


Figure 6. Intervals for the BT coefficients as a function of the number of samples and the number of models M .

this cannot be evaluated on real data, so we run a simulation. A vector of BT coefficients is drawn, with each coordinate sampled i.i.d. from a distribution $\text{beta}(1/\gamma, 1/\gamma)$; we take $\gamma = 2$ in Figure 6 (and we vary γ in Appendix A). Given these coefficients, a dataset is synthesized, and the coverage and average width are computed for each of 20 trials. The results can be seen in Figure 6 for the uncorrected intervals. The coverage of the intervals behaves as expected, centering around $1 - \alpha$, regardless of the number of models. Meanwhile, the more models are included, the larger the intervals become.

Evaluating the active sampling rule. Next, we discuss the evaluation of our active sampling rule as Equation (9) for win matrix estimation. We evaluate this sampling rule by taking the best fit BT coefficients to our 213,576 point sized holdout set, and then sampling from that distribution using our active sampling algorithm. The results are displayed in Figure 7. It is hard to tell by looking at plots, but the improvement is substantial: To estimate θ^* to a precision of 0.2, random needs 6,800 samples and adaptive needs 4,400 samples; meanwhile to estimate the score to a precision of 0.3, random needs 17,200 samples and adaptive needs 16,400 samples. Thus, the random baseline requires 54% and 5% more data to achieve the same level of precision, respectively. One can see from the plots in Figure 7 that these results are not cherry-picked: the sample-efficiency of our method is better at all values on the horizontal axis.

7.2. Anomalous Users Detection

We evaluate the outlier detection method in Section 5.1. We construct the evaluation set by manually identifying 25 anomalous users whose inputs are highly repetitive or meaningless (e.g., asking “hi” for 100 times or inputting garbled texts). We randomly sample 25 normal users with

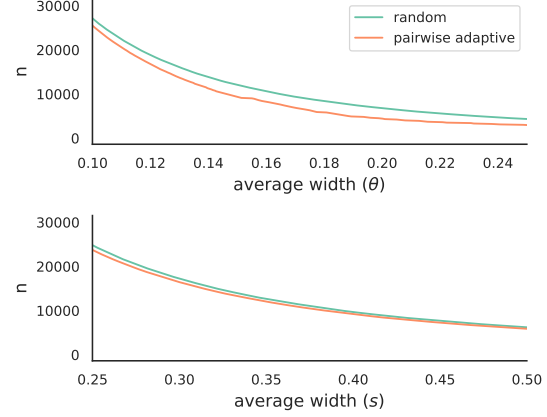


Figure 7. Interval widths on the win matrix (upper figure) and on the BT coefficients (lower figure) as a function of the number of samples, for random sampling and also adaptive sampling. Improvements from adaptive sampling can be seen in both cases, although they are more subtle on the scale of the score.

Table 5. Confusion matrix of different α . “Pred.” means predicted. Positive means anomalous and negative means normal.

$\alpha = 0.1$	Pred. Positive	Pred. Negative
Actual Positive	13/14	12/36
Actual Negative	1/14	24/36
$\alpha = 0.3$	Pred. Positive	Pred. Negative
Actual Positive	21/29	4/21
Actual Negative	8/29	17/21

at least 50 votes, and inspect their input prompts to ensure no abnormal behaviors. As mentioned in Section 5.1, per user we compute five M_j and identify the user as anomalous if $M_j \geq \chi_{2j, 1-\alpha/5}^2$. We present results of two different α (i.e., the significance level) in Table 5. We find the detection method effective (e.g., reaching 90% true positive and 60-70% true negative rate). We inspect the false negative errors and find those are from users do not always behave abnormally, making them harder to detect.

8. Discussion

Limitations. Although our user base is extensive, we anticipate that it will primarily consist of LLM hobbyists and researchers who are eager to experiment with and evaluate the latest LLMs. This inclination may result in a biased distribution of users. Additionally, despite the wide array of topics encompassed by the prompts discussed in previous sections, the data predominantly comes from our online chat interface. This source might not accurately reflect the real-world usage of LLMs in production environments or specialized domains, potentially leading to a skewed prompt distribution. Moreover, our study concentrates on assessing the helpfulness of LLMs but overlooks their safety aspects. We recognize the possibility and necessity of a parallel